SDG DIGITAL ASSIGNMENT - UNIT 2

UNIT 2

Q. Explain the role of hidden layers in a multilayer perceptron used to forecast urban air pollution levels from traffic density and weather variables. (SDG 11 – Sustainable Cities and Communities)

Urban air pollution (measured through indicators such as PM2.5, PM10, NOx, and CO2 concentration) is affected by a combination of factors including traffic density, wind speed, temperature, humidity, and time of day. These relationships are highly non-linear, meaning simple linear models cannot capture them accurately. A Multilayer Perceptron (MLP), a type of feed-forward Artificial Neural Network, is commonly used for this forecasting task because its hidden layers allow it to learn these complex, non-linear relationships. This directly supports SDG 11: Sustainable Cities and Communities by enabling accurate, timely warnings and planning for cleaner urban air.

1. Structure of the MLP for This Problem

Input Layer: Receives the raw input features — traffic density, vehicle speed/count, temperature, humidity, wind speed, atmospheric pressure, and time-of-day indicators.

Hidden Layer(s): One or more intermediate layers of neurons positioned between the input and output layers.

Output Layer: Produces the predicted pollution level (e.g., forecasted PM2.5 concentration for the next hour).

2. Role of Hidden Layers

Learning Non-Linear Relationships: Air pollution does not increase in direct proportion to traffic density alone — factors such as wind dispersing pollutants or humidity trapping particles create complex interactions. Hidden layers, using non-linear activation functions (e.g., ReLU, Sigmoid, or Tanh), allow the network to model these curved, non-linear relationships that a simple linear model cannot capture.

Feature Transformation and Abstraction: Each neuron in a hidden layer combines the inputs with learned weights and biases, transforming raw variables into new, more abstract representations. For example, an early hidden layer might learn a combined 'traffic-weather interaction' pattern (e.g., high traffic during low wind speed), which is more informative for prediction than either variable alone.

Hierarchical Pattern Learning (in deeper networks): When multiple hidden layers are stacked, earlier layers tend to learn simpler, low-level combinations (e.g., traffic + temperature), while deeper layers combine these into higher-level patterns (e.g., conditions strongly associated with pollution spikes during rush hour in low-wind, high-humidity weather).

Capturing Interactions Between Traffic and Weather Variables: Hidden layers allow the network to learn interaction effects — for instance, the impact of heavy traffic on pollution levels changes depending on whether wind speed is high (pollutants disperse) or low (pollutants accumulate). Hidden neurons can specialize in recognizing such combined conditions.

Dimensionality and Representation Learning: Hidden layers project the input features into a new representation space where patterns relevant to pollution levels are more linearly separable, making it easier for the output layer to produce an accurate final prediction.

Improving Generalization Through Depth and Width: The number of hidden layers (depth) and neurons per layer (width) control the model's capacity to learn complex patterns. Too few hidden neurons may cause underfitting (the model cannot capture the pollution-traffic-weather relationship well), while too many may cause overfitting (the model memorizes training data and performs poorly on new, unseen data). This is usually tuned using validation data.

3. How the Network Learns (Training Process)

Forward Propagation: Input data (traffic, weather values) passes through the hidden layers, where weighted sums and activation functions transform it, ultimately producing a predicted pollution value.

Error Calculation: The predicted value is compared with the actual recorded pollution level using a loss function (e.g., Mean Squared Error).

Backpropagation: The error is propagated backward through the hidden layers, and the weights and biases of each neuron are adjusted (using gradient descent) to reduce prediction error.

Iteration: This process repeats over many epochs until the network accurately predicts pollution levels from traffic and weather inputs.

4. Why Hidden Layers Are Essential Here

Without hidden layers, an MLP would essentially become a linear regression model, only capable of representing straight-line relationships between traffic density, weather variables, and pollution. Since real-world air quality depends on complex, interacting, and non-linear environmental processes, hidden layers give the network the computational power needed to approximate these relationships accurately.

5. Real-World Impact (Linking to SDG 11)

Enables early-warning systems that alert city authorities and citizens about expected pollution spikes

Supports smart traffic management decisions (e.g., rerouting or restricting traffic during high-risk conditions)

Assists urban planners in designing greener, low-emission zones based on predicted pollution patterns

Helps public health agencies issue timely advisories, reducing exposure-related health risks

Conclusion — Unit 2

In a Multilayer Perceptron used to forecast urban air pollution, hidden layers play the central role of learning the non-linear, interactive relationships between traffic density and weather variables that determine pollution levels. By transforming raw inputs into abstract, meaningful representations through weighted connections and non-linear activation functions, hidden layers enable the network to make accurate forecasts that support cleaner, safer, and more sustainable cities — directly contributing to SDG 11: Sustainable Cities and Communities.

— End of Assignment —